Unit 1: Using Objects and Methods
Test Review

Name: **KEY**

Multiple Choice

C 1) Consider the following code segment.

```
System.out.print("turtle ");
System.out.println("snake ");
System.out.println("dog ");
System.out.print("cat ");
```

What is printed as a result of executing the code segment?

(A) turtle
snake
dog cat

(B) turtle
snake
dog
cat

(C) turtle snake
dog
cat

(D) turtle snake
dog cat

C 2) Consider the following code segment.

```
int x = 3;
int y = /* missing expression */
x = 1 + 2 * y;
System.out.print(x);
System.out.println(y);
```

Which of the following can be used as a replacement for `/* missing expression */` so that the code segment prints 94?

(A) `3;`
(B) `x - 1;`
(C) `x + 1;`
(D) `1 - 2 * x;`

At the end of the code, x should be 9 and y should be 4.
Test each expression to see that D is the only one that gives us y = 4.

B 3) Consider the following code segment.

```
int x = 10;
int y = 20;
/* missing code */
System.out.print(top / bottom);
```

Which of the following replacements for `/* missing code */` will cause an `ArithmeticException` to occur?

I.
```
int top = x - y;
int bottom = y - x;
```

II.
```
int top = 2 * x;
int bottom = y - top;
```

III.
```
int top = x + y;
int bottom = 2 * top;
```

(A) I only
(B) II only
(C) I and II only
(D) II and III only

This error will occur when bottom = 0.
I. top will be –10 and bottom will be 10, and this is not an error.
II. top will be 20 and bottom will be 0, this will cause an error.
III. top will be 30 and bottom will be 60, and this is not an error.

D 4) Which of the following expressions will evaluate to 1?

I. `2 / 5 % 3`
II. `2 / (5 % 3)`
III. `2 / 5 + 1`

2 / 5 % 3	2 / (5 % 3)	2 / 5 + 1
0 % 3	2 / 2	0 + 1
0	1	1

(A) I only
(B) II only
(C) I and II only
(D) II and III only

A 5) Consider the following code.

```
int j = 12;
int k = 6;
j += 2;
k += j;
System.out.print(j + " " + k);
```

j	k
~~12~~	~~6~~
14	20

What is printed when the code segment executes?

(A) 14 20
(B) 14 18
(C) 8 12
(D) 12 6

D 6) Consider the following code segment.

```
int a = 10;
double b = 10.7;
double c = a + b;
int d = a + c;
System.out.println(c + " " + d);
```

a	b	c	d
10	10.7	20.7	ERROR

(A) 20 20
(B) 20.0 30
(C) 20.7 31
(D) Nothing will be printed because of an error

B 7) Which of the following statements will **NOT** result in a compiler error?

I. `int x = (double) 4.5;`
II. `int x = (int) 4.5;`
III. `int x = 4.5;`

For I and III, you are trying to fit a double into an int, and that will result in an error.
II is type-casting first, so you can fit it.

(A) I only
(B) II only
(C) III only
(D) I and II only

C 8) Given the following initializations,

```
double x = 5;
double y = 2;
```

What is the value of `m` after the following statement is executed?

```
int m = (int)(x + x / y - x * y - x / (10 * y));
```

(A) –1
(B) –0.75
(C) –2
(D) 0

D 9) What is the output of the following code?

```
int a = 1, b = 2, c = 3;
a += b + c; //a = 1 + 2 + 3 = 6
b += a + c; //b = 2 + 6 + 3 = 11
c += a + b; //c = 3 + 6 + 11 = 20
System.out.println(a + " " + b + " " + c);
```

(A) 3 3 4
(B) 5 4 3
(C) 5 8 13
(D) 6 11 20

D 10) What output is produced by the following line of code?

```
int myNumber = 4;
System.out.println("myNumber/3 = 1");
System.out.println(myNumber/3);
```

(A) 1.333333333333 = 1
1.333333333333

(B) 1 = 1
1

(C) myNumber/3 = 1
1.333333333333

(D) myNumber/3 = 1
1

B 11) What is the output of the following code?

```
String s = "ONION";
String first = s.substring(1,5);
String other = first.substring(1,4);
String last = other.substring(0,2);
System.out.println(last);
```

s = “ONION”
first = “NION”
other = “ION”
last = “IO”

(A) I
(B) IO
(C) ION
(D) ONI

C 12) Consider the following code segment.

```
String s = "goodbye";
System.out.println(s.substring(s.substring(s.length() - s.indexOf("d")).length()));
```

What is printed as a result of executing the code segment?

(A) bye
(B) 3
(C) dbye
(D) odbye

```
s.length() is 7
s.indexOf("d") is 3
s.substring(s.substring(7 - 3).length())
s.substring("bye".length())
s.substring(3)
= dbye
```

D 13) Consider the following declaration

```
Tool thing = new Tool();
```

Which of the following best describes the situation?

(A) `Tool` and `thing` are both classes
(B) `Tool` is an object of the `thing` class type
(C) `thing` is a primitive value of the `Tool` type
(D) `thing` is an instance of the `Tool` class type

B 14) Which of the following correctly gives random numbers between –10 and 10, inclusive, to the variable `n`?

(A) `int n = (int) (Math.random() * 11) - 20;`
(B) `int n = (int) (Math.random() * 21) - 10;`
(C) `int n = (int) (Math.random() * 10) - 21;`
(D) `int n = (int) (Math.random() * 20) - 10;`

–10 and 10 has a range of 21 and a min value of –10
(int)(Math.random() * range) + min

A 15) Consider the following code.

```
String q = "power";
String r = "brown";
System.out.println(q.indexOf(r.substring(3, 4)));
```

What is the output?

r.substring(3, 4) is w
q.indexOf("w") is 2

(A) 2
(B) 1
(C) 3
(D) w

C 16) A student has created a `Book` class. The class contains variables to represent the following:

- An `int` variable called `pages` to represent the number of pages
- A `boolean` variable called `isHardcover` to indicate whether or not the book is hardcover
- The object `story` will be declared as type `Book`.

Which of the following descriptions is accurate?

(A) An instance of the `story` class is `Book`.
(B) An instance of the `Book` object is `story`.
(C) An attribute of the `story` object is `isHardcover`.
(D) An attribute of the `Book` instance is `story`.

D 17) Consider the following class declaration.

```
public class Thing {
     private String color;

     public Thing() {
          color = "Blue";
     }

     public Thing(String c) {
          color = c;
     }
}
```

Which of the following code segments, when appearing in a class other than `Thing`, will NOT compile correctly?

(A) `Thing someThing = new Thing("Green");`
(B) `Thing someThing = new Thing();`
(C) `Thing someThing;`
(D)
```
Thing someThing;
something = Thing();
```

D 18) Consider the following code segment.

```
String word = "September";
String str1 = word.substring(0, 3);
String str2 = word.substring(word.length() - 3);
System.out.println(str1 + str2);
```

What is printed when the code segment is executed?

str1 = “Sep”
str2 = “ber”

(A) epber
(B) Sepbe
(C) Seper
(D) Sepber

B 19) Consider the following methods.

```
public void trial(){
      int a = 10;
      int b = 5;
      doublevalues(a,b);
      System.out.print(b);
      System.out.print(a);
}
public void doublevalues(int c, int d){
      c = c * 2;
      d = d * 2;
      System.out.print(c);
      System.out.print(d);
}
```

a = 10, b = 5
c = 10, d = 5
c = 20, d = 10
2010 first gets printed
510 then gets printed

What is printed as the result of the call to `trial()`?

(A) 2010105
(B) 2010510
(C) 20102010
(D) 20101020

D 20) Consider the following class declaration.

```
public class VetRecord {
      private String name;
      private int age;
      private int weight;
      private boolean needsVaccine;

      public VetRecord(String nameP, int ageP, int weightP, boolean needsVaccineP) {
            name = nameP;
            age = ageP;
            weight = weightP;
            needsVaccine = needsVaccineP;
      }

      public VetRecord(String nameP, int ageP, int weightP) {
            name = nameP;
            age = ageP;
            weight = weightP;
            needsVaccine = true;

      }
}
```

A new constructor is to be added to the `VetRecord` class. Which of the following is NOT a possible header for the new constructor?

(A) `public VetRecord(int ageP, boolean needsVaccine)`
(B) `public VetRecord(String nameP, int ageP)`
(C) `public VetRecord(String nameP, boolean needsVaccine)`
(D) `public VetRecord(String nameP, int weightP, int ageP)`

Free Response

21) Give 4 code examples of how to increment the integer `j` by 1.

```
j++        ++j        j += 1          j = j + 1
```

22)

a) For the following, give the range of possible values that can be displayed.	b) For the following, write a statement that generates a random number in the specified range (inclusive).

`Math.random() * 8`
`0 ≤ x < 8`
 	
`(int)(Math.random() * 20)`
`0 ≤ x ≤ 19`
 	
`(int)(Math.random() * 20) + 1`
`1 ≤ x ≤ 20`
 	
`(int)(Math.random() * 3 + 10)`
`10 ≤ x ≤ 12`
 	
`(int)(Math.random() * 100) + 16`
`16 ≤ x ≤ 115`
	0 to 10
	`(int)(Math.random()) *  11)`
	
	1 to 500
	`(int)(Math.random() * 500) + 1`
	
	10 to 15
	`(int)(Math.random() * 6) + 10`
	
	25 to 50
	`(int)(Math.random() * 26) + 25`
	
	–10 to 15
	`(int)(Math.random() * 26) – 10`

23) Consider the following initialization.

```
String name = "Godzilla";
```

What are the values of the following code segments?

`name.length()`
 	
`8`
	`name.substring(4)`
	
	`illa`
		`name.substring(0,3)`
		
		`God`

`name.substring(0,0)`
 	
`empty string`
	`name.indexOf("i")`
	
	`4`
		`name.indexOf("il")`
		
		`4`

`name.indexOf("za")`
 	
`-1`
	`name.compareTo("D")`
	
	`positive`
		`name.compareTo("Gt")`
		
		`negative`

24) Write a class called `Sphere`. This class should have the following:

- one `double` instance variable that represents the radius
- a default constructor that sets the radius to 1
- a parameter constructor that sets the radius to a passed value
- a method that returns the surface area using the formula $4\pi r^2$, where r represents the radius of the sphere
- a method that returns the volume using the formula $\frac{4}{3}\pi r^3$, where r represents the radius of the sphere

```
public class Sphere
{
	private double radius;

	public Sphere()
	{
		radius = 1.0;
	}

	public Sphere(double rad)
	{
		radius = rad;
	}

	public double surfaceArea()
	{
		return 4 * Math.PI * Math.pow(radius, 2);
	}

	public double volume()
	{
		return 4.0 / 3 * Math.PI * Math.pow(radius, 3);
	}
}
```